

Automated Polarized Sunlight-Blocking Film System

By Talal Alshamalial, G'avonn Jackson, Colton Miller, Kayla Pack, Samruddhi Patil

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Electrical & Computer Science Engineering, The University of Toledo
Faculty Advisors: Dr. Raghav Khanna, Dr. Daniel Georgiev
Instructor: Dr. Ashish Kharel
TA: Mehzabien Lqbal

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Problem Statement

People have long wished for a more convenient solution to both reducing the glare and heat coming into their homes. Homes with windows that face the sun face the complications of excess sunlight increasing the amount of light as well as heat that enters their homes. An added factor is that window-related heat loss is the largest contributor to cooling bills in the average household.

There are several solutions for these problems currently including traditional style window blinds and/or curtains, single-layer polarizing film (or tint), or smart windows. Blinds can block the light, but they also block the view people have of the outside of their homes. Smart glass solutions exist, but they are expensive for the average consumer as well as use a high amount of electricity. The automated polarizing window aims to find a solution to these problems by automatically adjusting the incoming light level, controlling it by an angle without reducing the view, while also reducing the amount of heat and light entering a home. The layers of polarized film in the system can adjust their angles in real time, which will keep the consumer space comfortable throughout the entire day. The design is geared to be affordable for the average consumer, making it a practical choice for homes and small buildings without the cost of a major renovation.

The automated polarizing window system is designed to offer many different angles for various levels of light adjustments.

The system has the capability to adjust polarization angles automatically based on the level of sunlight entering the home.

The system is also capable of allowing the user to turn off the automatic adjustment when the need to save power or set the light to be at a fixed level.

Furthermore, the frame is easy for the average user to both install and use while being energy efficient.

This product is more inexpensive and versatile than its smart-glass counterparts.

Finally, the product keeps the visibility to the outside of the home intact, while physical blinds block the view.

We were able to adapt our marketing requirements into specific engineering requirements that helped us with the overall design of our product. We were able to take these requirements and piece together the different aspects that our project needed to achieve.

1. Light Levels: The polarized film system should provide at least six different light levels to give users precise lighting control for brightness and visibility. This selected range ensures the ability to find comfort in a wide variety of settings, optimizing light preferences.
2. Angles Range: The polarized film system should provide light levels with angles from 0° to 90° to achieve full degree of light control. This range aids in providing visual comfort to users and reduces heat buildup from direct sunlight exposure.
3. Light Sensor: The light sensor should measure a range of at least 256 different light levels to effectively measure light changing conditions. This range allows for automatic adjustment for illumination and cost efficiency.
4. Adjustment Time: The film system should adjust the film within 120 seconds (and not more than a worst-case value of 180 seconds). This fast response time makes sure that users are provided with a seamless light transition without delay.
5. Dimensions: The frame should fit standard windows 24-40 inches wide and 36-62 inches high. This range will provide compatibility with most household and office windows. Any other dimensions would require a custom-built frame.
6. Install Time: The system should be installed within 30 minutes by an able-bodied/average person using little to no tools. This quick installation will increase user satisfaction and help reduce setup costs.
7. Cost: The product cost should be less than 75% of the price of smart glass (electrochromic glass). This will provide affordability and high performance, and a reasonable price point. This will give a competitive edge from pre-existing products already on the market.

By using these listed requirements, both marketing and engineering, we were able to adapt our design and development process to be able to suit the needs of a consumer, creating a production ready design.

System Design

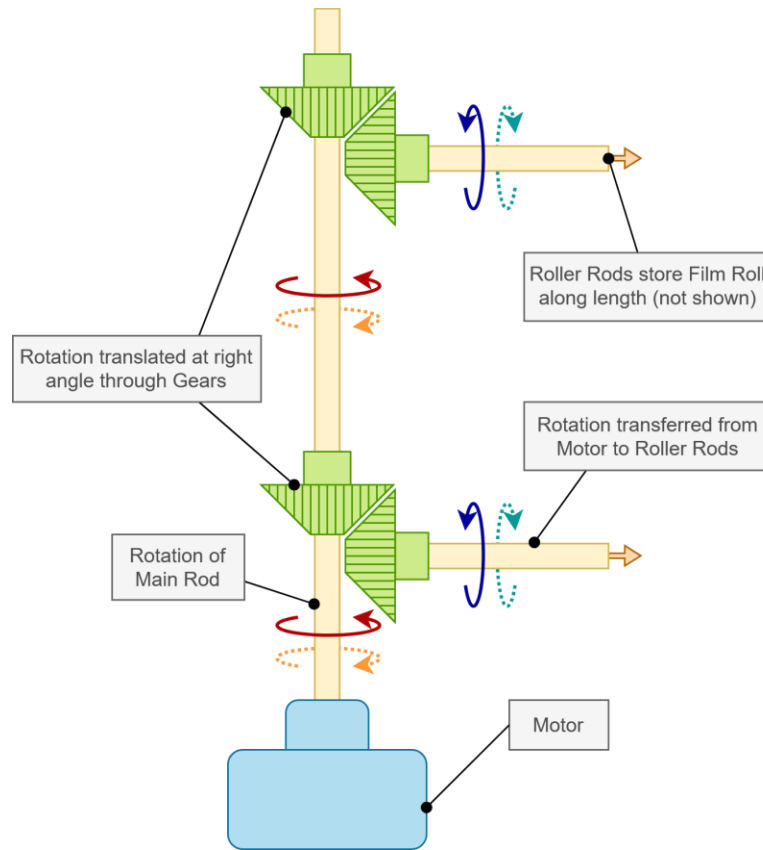
Software

The software component of our project was limited to programming control logic for the Arduino and motor. It managed sensor inputs from the 2 light sensors and Bluetooth module while controlling the motor via the motor driver.

The Arduino Uno Rev3 was controlled via Arduino Sketch (C/C++). The primary component that is controlled by the Arduino is the stepper motor. The stepper motor is controlled by two light sensors when the program is running in “Auto” mode. The film is broken into segments; each segment has a measured and tested step count that the motor must move to go from one segment to the next. Using the light sensors, the Arduino can automatically select the desired segment and will turn the motor the needed number of steps to reach the desired point. We also have a manual mode that allows the user to select any individual segment using Bluetooth commands from their phone. This works in the same way where the program tracks the current location and will calculate exactly where to move the motor depending on input.

Frame

The structural frame is one of the most core parts of our automated polarizing window system because it acts as the main support structure to house our critical system components. It holds and organizes all the key components; this includes the motion systems (rods/polarizing + clear film, gears/couplers, bearings/holders) as well as the electrical control systems (driver, motor, Arduino, light sensors, and Bluetooth module). The main purpose of the frame is to keep everything aligned and stable so the system can function properly as a whole. Without this alignment, the rotating parts would not move smoothly, causing a serious issue with film rotation.



(Rotational diagram of rods connected to attached gears)

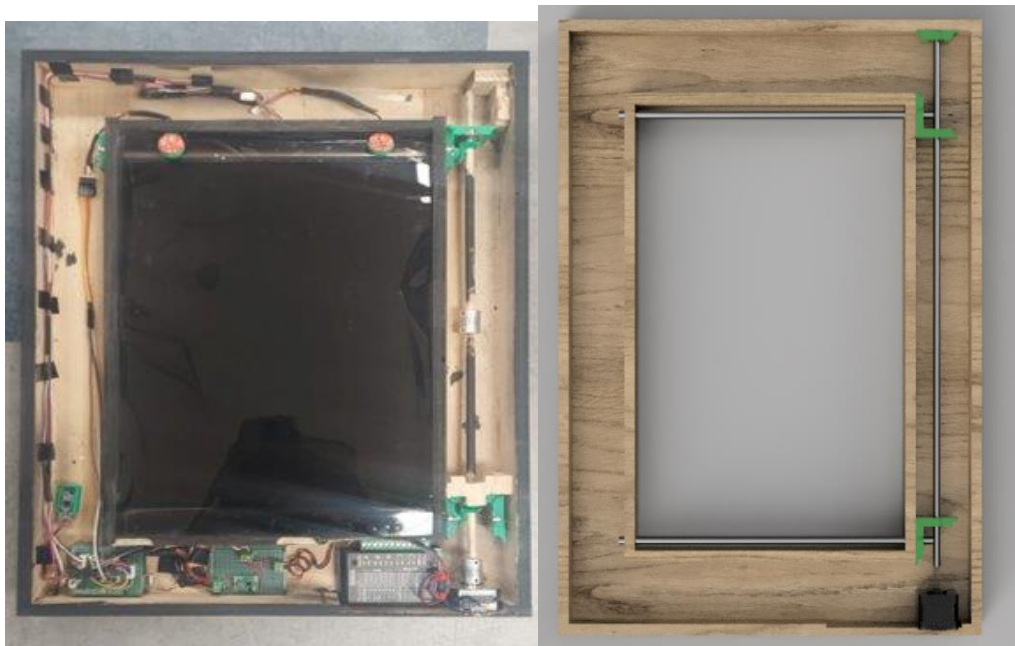
The frame design is based on plywood, which was selected because it is strong, lightweight, cost-effective, and is easy to cut and assemble for prototyping. We constructed the frame by cutting the plywood into individual sections and assembling the sections using screws and a nail gun to ensure structural support at key points. This approach allowed us to create a better overall structure while still being able to make adjustments during the building process. The layout was carefully planned to support both mechanical and electrical placement, including mounting points for our rods, slit openings for the sensors, and routing space for wiring and power input.

Additional slits were drilled into the frame using a Dremel to support metal clips, which were bent and cut to help maintain tension between the frame and the window. This

system was used as a mounting method to securely attach the window to the frame while also allowing slight flexibility to prevent stress on the materials. This helped keep the polarizing films properly positioned and prevent unwanted movement during operation.

For the electrical integration, we specifically allocated space within the frame to mount the Arduino, Bluetooth, motor, driver, and light sensors. Sensor placement was designed to ensure accurate light detection while remaining protected to support the polarizing system. This improves the system's ability to respond to real world lighting conditions more effectively.

A dedicated opening was also included below the frame to provide a clean and accessible power input for the electrical components. Proper wiring management was used within the shell of the frame, with wires routed along the shell structure of the frame to prevent interference with moving parts such as rods and gears. This not only improves safety by reducing the risk of loose or exposed wiring, but also keeps the system organized and easier to maintain or troubleshoot in the future. Overall, the frame design supports both mechanical stability and reliable electrical integration, allowing the system to operate smoothly and consistently.



(Finalized and initial version of our integrated frame)

Overall, the frame acts as a structural support system that interfaces both mechanical and electrical systems. It ensures that all components work together in a stable and controlled environment, helping to provide smooth rotation of the polarizing films while keeping the system secure. In addition, the frame also helps reduce any unwanted movement from the rotational movement, vibration, and any misalignment that could impact system performance over time. It also provides organized mounting for electrical components, which ensures a safe placement for all connections. By keeping both the mechanical and electrical elements integrated and separated, this improves functionality and decreases hazardous risk that could come from wires getting caught on the mechanical components.

Polarizing film

The project's core functionality is realized through and based around linear polarizing film. The idea was to have multiple overlapping layers of linear polarizer film, such that the combination of multiple layers at different polarization angles could give a filtered output of light. This filtered output of light can be very finely controlled by picking the correct combination of film angles. This initial idea was planned to have two separate rollers with different variance of angles, as well as the base layer polarizer which remained at a static angle. Following further revision and fine-tuning of the design, however, we settled on a simpler, single-roller design, which consisted of only a single roller and the base layer of film. This streamlined the design and assembly process, albeit losing some granularity in the control of output light level. This very importantly cut down on costs significantly, seeing as the most expensive ordered material for the project was the polarizing film and the associated import fees and tariffs (despite not being particularly difficult to manufacture), and made up over half our allocated budget for the project at ~\$170.00.

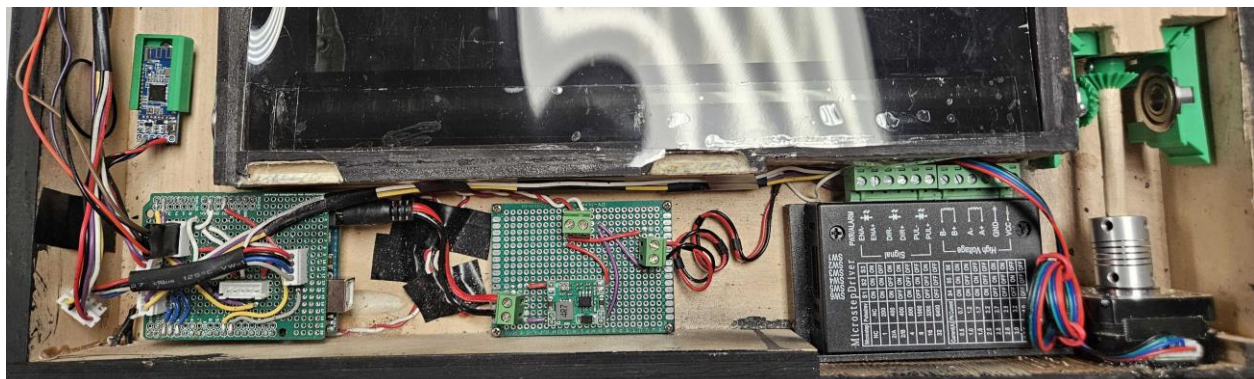
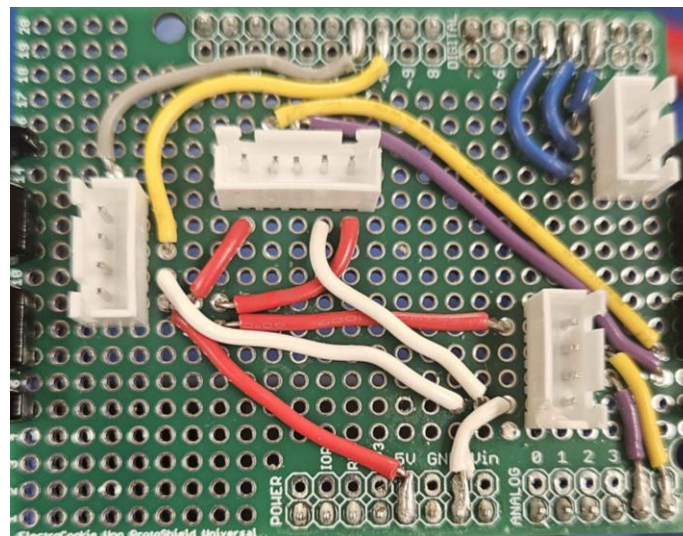
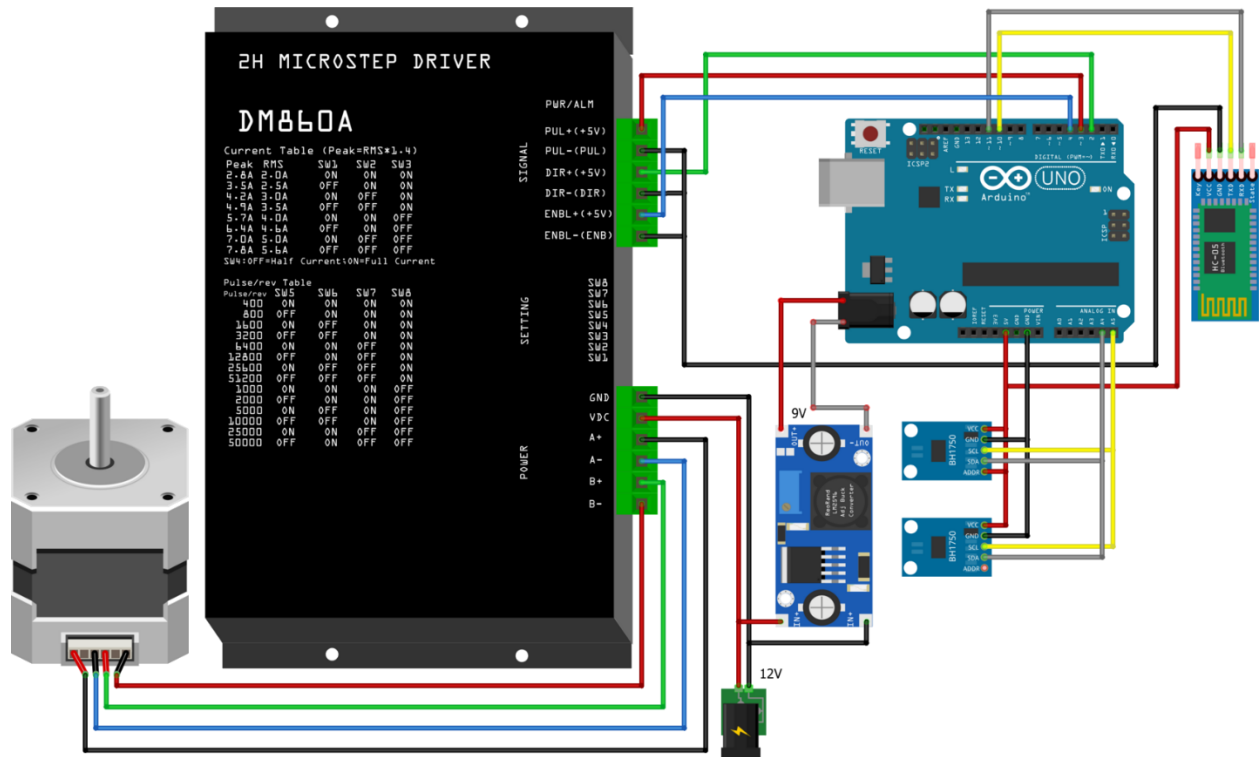
We decided to pick angles corresponding to 20% increments in output light levels, i.e. the light intensity that came through the static sheet of film was then divided into 5 different film options scaling by a factor of 0.0, 0.2, 0.4, 0.6, 0.8 and 1. This ended up being the angles

Electrical

The electrical system functionality of our project covers a vast array of electrical components that we used. Our window features a plug-in functionality using a 12VDC adapter from an electrical outlet. From here, we had to distribute power to a 12VDC circuit as well as a 9VDC circuit. We utilized a buck converter for this power conversion, to drop the 12VDC down to 9VDC. This allowed us to have a single power source that would be able to supply power to components that required a range of voltages. The 12VDC circuit provided power to our motor and motor driver while the 9VDC circuit provided power to the

Arduino, Bluetooth module, and light sensors. We utilized a stepper motor with a driver to control the precise steps that the motor needed to take to accurately turn the roll of film to the correct segment. An Arduino was used as the main microcontroller that code was written to for these precise calculations. We were able to integrate a Bluetooth module to be able to connect with the user's phone to manually control the polarization level. We were then able to run wires across the frame that connected the light sensors to the Arduino for precise sunlight measurement.

The circuit design started from a concept on a breadboard until we were able to successfully develop a permanent solution within our frame design. We drilled a hole in the frame where the power supply would be able to plug into the components on the inside of the frame. This power source is connected to a circuit that includes our buck converter. The buck converter circuit then branched power to both the driver and the Arduino. The driver received power from the power distribution circuit and provided inputs to the stepper motor itself. The Arduino was powered directly from the circuit containing the buck converter. A proto-shield was affixed to the Arduino which contained a circuit that provided jumpers to the various modules that we used. We used JST-XH connectors on the proto-shield along with crimped wires that ran the connections from the Arduino. Wires were traced on the proto-shield to create connectivity without the use of dupont jumper cables. This circuit provides power and connectivity for the Bluetooth module which we mounted on the inside wall of the frame. From this circuit we provided connections to both of our light sensors which provided input signals to the Arduino about the level of light that the lux sensor was receiving. The proto-shield on the Arduino also provided an output signal which was wired to the motor driver to provide precise stepping instructions.



Testing & Validation

Testing was an ongoing requirement through the development of our product. Our first round of testing consisted of writing the code and being able to connect the Arduino to the other components in our design. The original circuit was built on a breadboard design with dupont connectors. We were able to test the basic functionality of the motor responding to the code that was written to the Arduino. From there we were able to add the light sensors to our initial circuit design and test that they were able to read the various light levels. Once the initial functionality of our circuit was concluded to be functional, we were able to begin the design and building of the frame that would house the circuit and the overall finished product.

Once the initial design of the frame was complete, we began to test the components as they were placed into the frame. While testing and ensuring things would fit, we had to backtrace the frame design to allow more room for certain components. In this, we tested the overall functionality by allowing the gears to spin the rods while they were in the frame. From here we were able to determine areas where the rods might catch on the frame and furthermore adjust our frame design to fit the spatial needs of our rods and rollers.

After ensuring that the pieces properly fit into the frame, and things were able to roll without catching on the frame, we were able to test our remote-control design of the motor. From the app, we were able to test each segment of our polarizer film and ensure that the motor turned the roller the proper number of times to reach the next segment. We had to fine tune our calculations from here to ensure that the motor turned the proper number of times to show the entirety of the adjusted light level.

Once we ensured that we could manually select the light percentage that we wanted to be allowed into the room, we were then able to begin the integration of our light sensors. We had to ensure that the code would properly read the value that was being sent from the light sensors and communicating it with the motor. We used various levels of brightness of light from various sources to ensure that the system would successfully read the percentage of light and adjust the rollers accordingly.

Alternative Designs

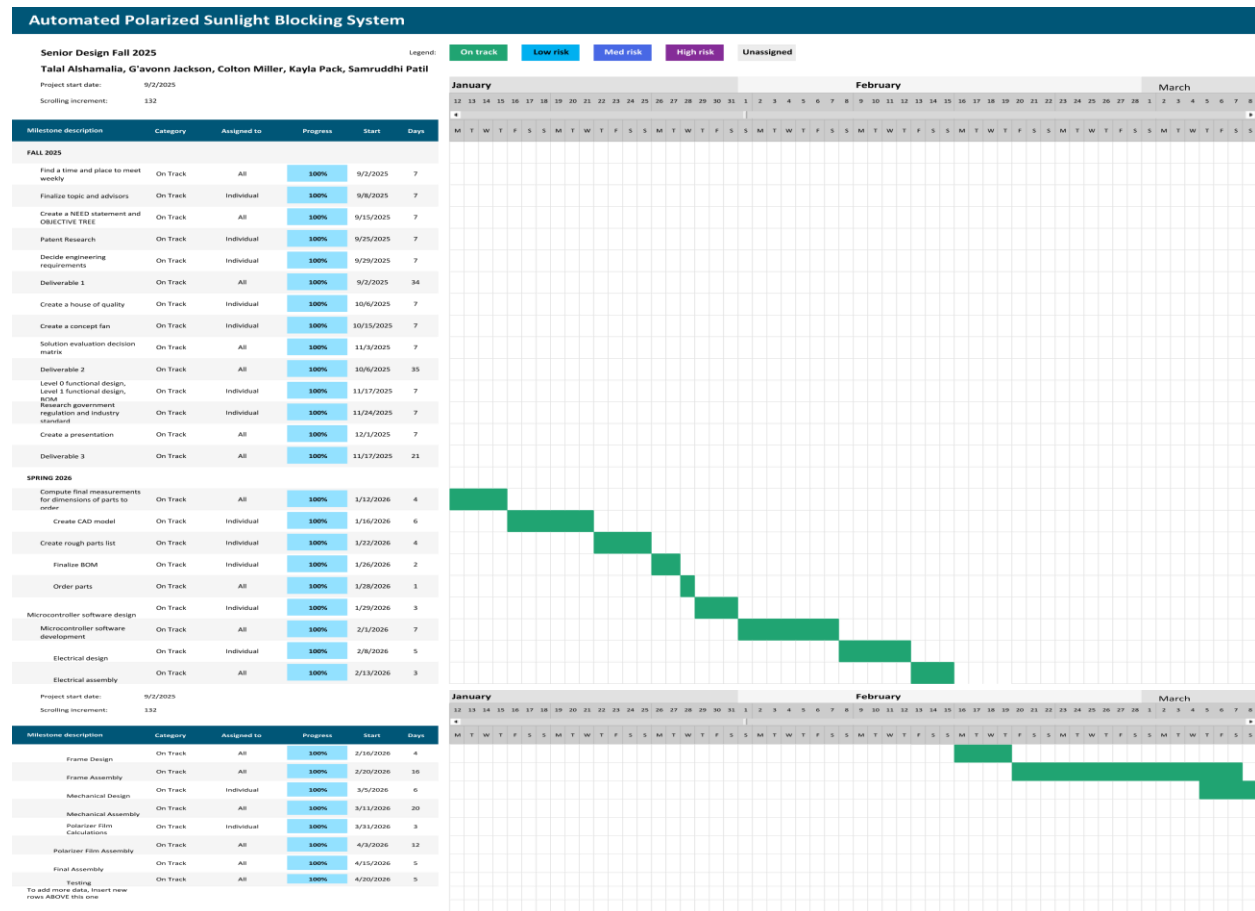
Alternative solutions include multiple rollers, multiple motors, and a circular window design. The use of multiple motors would allow more stable control over the gears, which in turn would allow the rods holding the polarizing film to rotate more smoothly. Having

multiple rolls of polarizer film would allow for more precise control over the angle of the polarizer, allowing a wider range of light levels to be selected from. A circular window would allow for a smoother transition of light levels, as one layer of the polarizer film would be able to rotate on its own.

We decided on a single motor design because while having a second motor would help to adjust the tension of the film, it was deemed unnecessary due to cost as well as the unnecessary need to add more components to the inside of the frame. Having two rolls of polarizer film would be clunky in a design and not make it appealing to the average consumer. Finally, a design using a circular window instead of a square window is not realistic as very few homes or small buildings would have these round windows that the frame could clip onto.

Our project settled on a design that included one motor, one roll of film, and a square design. We chose this design for multiple reasons. We decided upon a rectangular window design, as the average consumer does not have circular windows in their homes. We wanted to make our product accessible and available for the wide majority of consumers. A single motor makes the production of the frame easier and more cost effective as does having only one roll of polarizer film. Introducing multiple rolls of polarizer film allows for more precise angle control. However, with the price of polarizer film, it is not a cost-effective method for our purpose. Having one roll of polarizer film already takes up a significant amount of space as well, making a second roll not having much room without an entire frame redesign. Redesigning the frame may make it not have as many capabilities of retrofitting any sort of window, making our final design option the best choice.

Cost Breakdown



We were able to successfully create and follow our Gantt chart throughout the course of the semester and utilized it heavily to ensure that we stayed on track for progress.

Part Number	Description	Quantity	Cost
ARD_A000066	Arduino Mega	1	\$27.60
MP1584EN	Buck Converter	1	\$2.05
DM860A	Motor Driver	1	\$11.89
17HE08-1004S	Motor	1	\$10.50
BH1750FVI	Light Sensor	2	\$11.99
-	Plywood	1	\$5.00
W1218BR-BX1	Window	1	\$29.00
-	Drywall	1	\$10.23
PT-4mil-JUAN	Clear Film	1	\$16.99
LS35CP	Polarizer Film	1	\$176.99
-	3D Printed Parts	1	\$3.00
-	Wires and Misc	1	\$40.00
HM10	Bluetooth Module	1	\$10.00
B09YV11ZY7	Rod Coupler	2	\$6.99
H20221231017	Bearings	6	\$12.49
Total			\$374.72

After totaling our bill of materials, we slightly exceeded our budget, ending at a total of \$374.72. Much of the plywood that we used in our project was provided to us free of charge, making a realistic end total closer to \$400.00. This number would be able to be reduced given an exact amount of polarizer film was ordered from a different supplier. Lead times on polarizer film as well as tariffs drove the price on the film higher than anticipated.

Regulatory Standards

There are several government regulations and standards that have are in place. ENERGY STAR is a performance criterion that places certification on a product. With this certification, homeowners and manufacturers can receive federal tax credits or other incentives. Furthermore, since the product uses a window film system, there are fire-rating and film-rating systems that must are considered in development. Dynamic polarization must ensure fire-safety compliance through local fire codes. This requirement must be met if the window changes optical properties. Further testing must test strength, safety compliance, and risk of breakage must not be affected by the product.

Several other codes sit in place to handle things relating to polarized windows and energy ratings. The National Fenestration Rating Council (NFRC) provides standardization for energy performance testing, certifications, and labeling for window additions. There is also

the International Energy Conservation Code which regulates the fact that window additions are to not make the performance of the window have a disadvantage over a standard window without additions. This code establishes the minimum design and construction requirements for energy efficiency.

There were several ethical issues that were considered in the design of the product. Several names for our product that could not be used, including: “SMART TINT”, “SMART CLING”, “SMART WINDOW TINT”, SMART GLASS”, and “VIEW”. A patent exists that covers some control and automation aspects of smart windows which cover a multi-zone tinting algorithm, remote monitoring, and building-system integrations. Another patent exists that regards transmittance-adjustable windows using polarizer pairs with actuators.

The design of our product is accessible for most, if not all, consumers. It eliminates the need to physically stand up and walk to your window if you want to adjust the amount of light being let into your room. It allows you remotely or sets a schedule for the light level adjustment. This can be either remotely controlled from an app, or it can be set on an automatic schedule that will adjust the incoming light level based on the percentage of sunlight that is directly hitting the window.

<https://patents.justia.com/assignee/sage-electrochromics-inc>
<https://patents.google.com/patent/US5164856>

Results

The design of the automatic polarizer window includes six segments which adjust to different angles of polarization. These angles of polarization represent real-world levels of sunlight and cover a range from early morning sunlight to direct and peak sunlight. These six segments cover a range of 0 to over 30,000 lux and can dim all light within this range. The first segment covers the light the average consumer would see in the early morning. Our tested range includes low daylight, cloudy daylight, strong daylight, partial direct sunlight, and direct sunlight.

In testing, the system showed increased performance with additional code changes and support implementation for the holders and rod adjustments. The motor with the coupled rod attachment was able to successfully rotate the polarizing film rods to our desired angles, showing that the mechanical design does support our system functionality. Additional testing of the light sensors allowed our system to respond to the different changes in light levels we exposed them to; this validated the effectiveness of the manual and automatic real time adjustments.

The results we obtained closely align with the original system requirements, mainly in terms of maintaining visibility while reducing glare and controlling the incoming light through the window, which was one the primary goal of our design. Additionally, the range of polarization angles used provided gradual transitions between different lighting controls.

While the system performed well overall, there were a few minor issues for the light sensors that slightly impacted system performance but were later fixed. Additionally, improved positioning was needed to better align the rods to help insure reduced slipping and tension loss. Overall, our results provide strong evidence that our systems design meets the intended objectives and demonstrates a viable solution for automated light control in residential settings.

Conclusions & Future Work

Overall, we achieved satisfactory results in all major system subsections. The mechanical aspects work well for our use case but have clear room for improvement in terms of both reliability and durability.

The electrical systems are sufficient and work properly for our project and are well designed overall. By using a DC power jack to provide power from a wall outlet, it is easy and safe to connect and disconnect power from the window as needed. By integrating multiple circuits on boards that we hand-soldered, we were able to create an excellent electrical component design.

The software is functional and completes all set requirements but is still more of a skeleton model. There are further smart features that could be implemented using Bluetooth connectivity such as the creation of an app. This would provide an improved user interface, and we could also provide more advanced logic to allow more user control.

Structurally and aesthetically, the window exceeded our initial goals. The wooden structure of the frame is sound and durable. After painting and installation of all parts, the overall package looks professional and is well designed for our use case.

Future Improvement

A meaningful future improvement would be a dual-roller system that would allow for significantly finer control over the specific light levels that the window lets through. By being able to combine multiple angles in multiple layers, it would be possible to achieve more than double the number of light levels. Our project has two light sensors, but a more complex array of sensors would allow for more advanced smart features. A possible addition and improvement to the project would be the addition of temperature or heat

sensors to the current two light sensors. This would allow users to select a desired amount of heat the passes through the window as well as making the light-based controls more precise.

Another future improvement could be the implementation of a dual-motor solution instead of a more complicated drive-shaft system for the rollers. This would allow for more simple control and increase the reliability of our product. It would also pose the major benefit of allowing automated tensioning of the film to ensure that it stays taut over time.

In conclusion, we were able to achieve full functionality of all components and meet all of our requirements, goals, and expectations. Our product has reached a stage of complete and functional implementation.